

| Type               | Content                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Seed Term Prompt   | data augmentation is used for Task                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Context            | data augmentation is an effective solution to data scarcity in low - resource scenarios. however, when applied to token-level tasks such as ner , data augmentation methods often suffer from token-label misalignment, which leads to unsatisfactory performance.                                                                                                                                                                                  |
| Semantic Neighbors | st and automatic speech recognition (asr), low-resource tagging tasks, end-to-end speech translation, neural online chats response selection, neural machine translation, semi-supervised ner, entity and context learning, semi-supervised setting, dependency parsing, low-resource machine translation, slot filling, dialog state tracking, visual question answering, visual question answering (vqa), low-resource neural machine translation |
| KG Neighbors       | nmt-based text normalization, task-oriented dialog systems, task-oriented dialogue system, low-resource languages (lrl), end-to-end speech translation, visual question answering (vqa), multiclass utterance classification, clinical semantic textual similarity, neural online chats response selection, context-aware neural machine translation                                                                                                |
| Citation Neighbors | Contextual Augmentation: Data Augmentation by Words with Paradigmatic Relations, An Analysis of Simple Data Augmentation for Named Entity Recognition, Data Augmentation for Low-Resource Neural Machine Translation, DAGA: Data Augmentation with a Generation Approach for Low-resource Tagging Tasks, EDA: Easy Data Augmentation Techniques for Boosting Performance on Text Classification Tasks                                               |
| Ground Truth       | ELM: Data Augmentation with Masked Entity Language Modeling for Low-Resource NER                                                                                                                                                                                                                                                                                                                                                                    |

Table 8: Example (from (Zhou et al., 2022)) of retrieved inspirations. Inspirations similar to ground truth are underlined.

| input                                                 | context                                                                                                                                                                                                                                                                                                                            | entity                        | output      | relation | rel_sent                                                                                                                                                                                                                                                                                                                                           | Is the output trivially overlap with the context | IE is of sufficient quality (not generic, correct) | context contains relevant information for target relation (Conservative filter - only flag cases where context is highly irrelevant) | Relation is a part of the main idea proposed by the paper |
|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|-------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| extractive text summarization is done by using Metric | transformer - based language models usually treat texts as linear sequences . however , most texts also have an inherent hierarchical structure , i.e. , parts of a text can be identified using their position in this hierarchy . in addition , section titles usually indicate the common topic of their respective sentences . | extractive text summarization | sota rouges | used for | We propose a novel approach to formulate , extract , encode and inject hierarchical structure information explicitly into an extractive summarization model based on a pre - trained , encoder - only Transformer language model ( HiStruct+ model ) , which improves SOTA ROUGes for extractive summarization on PubMed and arXiv substantially . |                                                  |                                                    |                                                                                                                                      |                                                           |

Figure 5: Gold subset annotation interface

anthology and PubMed papers written in English, which might alienate readers who have been historically underrepresented in the NLP/biochemical domains.

## E.2 Data Collection

We collect 67,408 ACL Anthology papers from 1952 to 2022 using Semantic Scholar Academic Graph API, under API license agreement<sup>26</sup>. We ensure our data collection procedure follows the Terms of Use at <https://allenai.org/terms>. According to the agreement, our dataset can only be used for non-commercial purposes. As mentioned in §4, we perform the human evaluation. All

annotators involved in human evaluation are voluntary participants with a fair wage. We further collect 5,708 PubMed papers from 1988 to 2024 using Entrez Programming Utilities API<sup>27</sup>. We follow their data usage guidelines<sup>28</sup>.

<sup>26</sup><https://api.semanticscholar.org/license/>

<sup>27</sup>[www.ncbi.nlm.nih.gov/books/NBK25501/](http://www.ncbi.nlm.nih.gov/books/NBK25501/)

<sup>28</sup>[www.ncbi.nlm.nih.gov/books/about/copyright/](http://www.ncbi.nlm.nih.gov/books/about/copyright/)

| Subset    | Challenging                       |                                   | Gold                              |                                   |
|-----------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
| Model     | R-L $\uparrow$                    | BERT $\uparrow$                   | R-L $\uparrow$                    | BERT $\uparrow$                   |
| GPT4ZS    | 0.120                             | 0.581                             | 0.130                             | 0.583                             |
| GPT4FS    | 0.143                             | 0.618                             | 0.151                             | 0.624                             |
| T5        | 0.223                             | <b>0.672<math>^\dagger</math></b> | 0.246                             | 0.685                             |
| GPT4FS+SN | 0.144                             | 0.620                             | 0.149                             | 0.627                             |
| GPT4FS+KG | 0.143                             | 0.619                             | 0.152                             | 0.626                             |
| GPT4FS+CT | 0.144                             | 0.617                             | 0.149                             | 0.622                             |
| T5+CL     | 0.225 $^\dagger$                  | 0.671 $^\dagger$                  | 0.251 $^\dagger$                  | 0.686 $^\dagger$                  |
| T5+SN+CL  | <b>0.228<math>^\dagger</math></b> | 0.671 $^\dagger$                  | <b>0.258<math>^\dagger</math></b> | 0.686 $^\dagger$                  |
| T5+KG+CL  | 0.223 $^\dagger$                  | 0.669                             | 0.248                             | 0.681 $^\dagger$                  |
| T5+CT+CL  | 0.225 $^\dagger$                  | 0.671 $^\dagger$                  | 0.250 $^\dagger$                  | <b>0.686<math>^\dagger</math></b> |

Table 9: Automatic evaluation results for the challenging and gold subsets. *CL* is a model with in-context contrastive augmentation. *SN* is a model with semantic inspirations. *KG* is a model with KG inspirations. *CT* is a model with citation inspirations. *R-L* denotes ROUGE-L. *BERT* denotes BERTscore with SciBERT as its encoder.  $^\dagger$  indicates that differences between models are not statistically significant ( $p \leq 0.05$ ) when compared to each other but are still significant when compared to the other models based on t-test.

| Type             | Content                                                                                                                                                                                             |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Seed Term Prompt | Irish language learning is done by using Method                                                                                                                                                     |
| Context          | Irish is a minority language which means that 12 learners have limited opportunities for exposure to the language, and additionally, there are also limited (digital) learning resources available. |

Table 10: Input for sample human annotation results

| Model       | Output                                                                                                                                                                                                                                                                                                                                                                                   | Label            |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| GPT3.5FS    | The use of Social Media in <b>Irish Language Learning</b> : A Case Study                                                                                                                                                                                                                                                                                                                 | <i>helpful</i>   |
| GPT3.5Retr  | One method that could be used for <b>Irish language learning</b> is computer assisted language learning (CALL)                                                                                                                                                                                                                                                                           | <i>unhelpful</i> |
| GPT3.5FS+CT | In that context, the most appropriate task for precision grammar would be the development of a part-of-speech tagger for Irish using finite-state morphology and constraint grammar disambiguation.                                                                                                                                                                                      | <i>unhelpful</i> |
| GPT3.5FS+KG | Cló: An <b>Irish Language Learning</b> Platform - Leveraging Digital Resources and Human Expertise                                                                                                                                                                                                                                                                                       | <i>unhelpful</i> |
| GPT4FS      | We propose using a technology-enhanced approach, such as computer-assisted language learning (CALL) or natural language processing (NLP) techniques, to create interactive and engaging <b>Irish language learning</b> resources, as these methods can provide personalized learning experiences and enhance learners' exposure to the language despite limited resources available.     | <i>helpful</i>   |
| GPT4FS+KG   | Exploiting Social Media for Irish Language Learning: An Analysis of Twitter Data. In this context, we use social media data, particularly from Twitter, as a method for <b>Irish language learning</b> , because it provides a rich source of authentic and diverse language examples that can be used to enhance learning opportunities for L2 learners in a minority language setting. | <i>helpful</i>   |
| T5          | In this paper, we present a novel machine translation approach to <b>Irish language learning</b> .                                                                                                                                                                                                                                                                                       | <i>unhelpful</i> |
| T5+SN+CL    | A conversational virtual environment for <b>Irish language learning</b>                                                                                                                                                                                                                                                                                                                  | <i>helpful</i>   |

Table 11: Sample human labels for idea-sentence generation results for seed term prompt “*Irish language learning is done by using Method*”, with context “*irish is a minority language ... limited digital learning resources available*”.

| Type          | Content                                                                                                                                                                                                                                           |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Seed Term     | ARO10                                                                                                                                                                                                                                             |
| Context       | Huangjiu is known for its unique aroma, primarily attributed to its high concentration of b-phenylethanol (ranging from 40 to 130 mg/L).                                                                                                          |
| Meditron      | We then selected six key genes involved in the phenylalanine metabolism pathway and predicted that their product flux may affect the synthesis of b-phenylethanol.                                                                                |
| Meditron + CT | We found that the key amino acid residue that controls the activity of <b>Aro10p</b> was not conserved in wine yeast strains, which may explain the lower b-phenylethanol production in wine fermentation compared with that in Chinese huangjiu. |
| Meditron + KG | Both target genes, <b>SSA1</b> and <b>ARO10</b> , were deleted using the CRISPR-Cas9 genome editing system.                                                                                                                                       |
| Meditron + SN | Herein, we report that the key barrier for b-phenylethanol production in Huangjiu is <b>ARO10</b> , the only bi-functional amino acid decarboxylase in <i>Saccharomyces cerevisiae</i> .                                                          |

Table 12: Input and idea-sentence generation results for seed gene “*ARO10*” in the biochemical domain.

| Annotator Pair | 1-2  | 1-3  | 1-4  | 1-5  | 1-6  |
|----------------|------|------|------|------|------|
| Agreement %    | 68.8 | 75.0 | 56.2 | 43.8 | 75.0 |

Table 13: Percent (%) of same labels from overlapped 10 human evaluation instances on each pair of annotators for Study I.

| Annotator Pair | 1-2  | 1-3  | 1-4  | 1-5  |
|----------------|------|------|------|------|
| Agreement %    | 92.5 | 93.3 | 87.5 | 90.0 |

Table 14: Percent (%) of same labels from overlapped 20 human evaluation instances on each pair of annotators for Study III. (1-3) has 60 shared questions. The rest of the pairs each share 40 questions.